

Lab Experiments

Set 18: Product Inventory Management

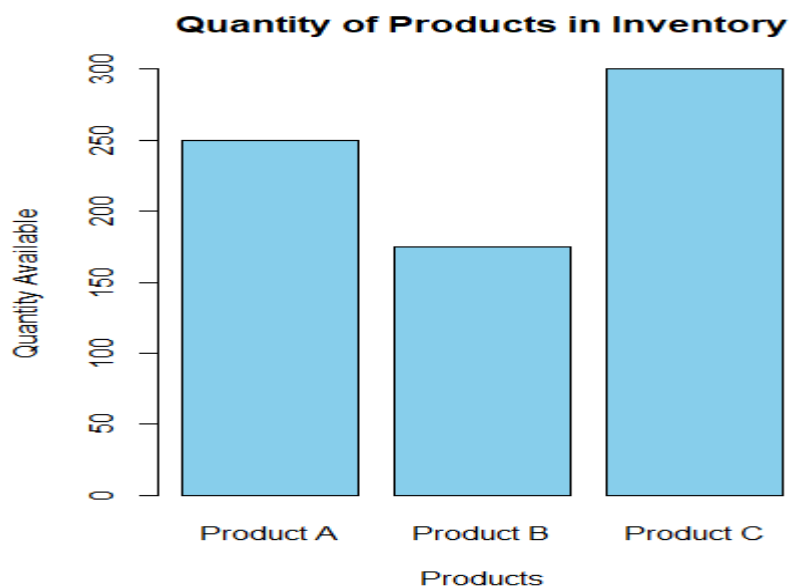
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

Program:

```
# Product Inventory Dataset
inventory <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  Quantity = c(250, 175, 300),
  Price = c(20, 15, 18)
)

# Bar Chart
barplot(inventory$Quantity,
        names.arg = inventory$Product,
        col = "skyblue",
        main = "Quantity of Products in Inventory",
        xlab = "Products",
        ylab = "Quantity Available")
```

Output:



2. Generate a stacked bar chart to show the quantity of each product within different product categories.

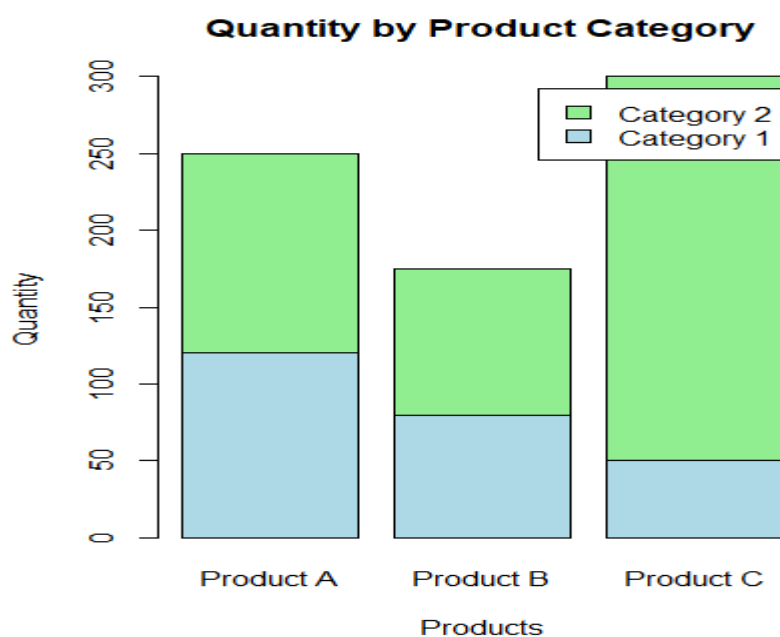
Program:

```
# Product Categories
categories <- matrix(c(
  120, 80, 50,
  130, 95, 250
), nrow = 2, byrow = TRUE)

colnames(categories) <- c("Product A", "Product B", "Product C")
rownames(categories) <- c("Category 1", "Category 2")

barplot(categories,
  col = c("lightblue", "lightgreen"),
  main = "Quantity by Product Category",
  xlab = "Products",
  ylab = "Quantity",
  legend.text = rownames(categories))
```

Output:



3. Build a table to show the inventory data for each product. Label the table elements.

Program:

```
inventory <- data.frame(  
  Product_ID = c(1,2,3),  
  Product_Name = c("Product A","Product B","Product C"),  
  Quantity = c(250,175,300),  
  Price = c(20,15,18)  
)
```

Inventory

Output:

```
3      3  Product C      300      18  
> inventory  
  Product_ID Product_Name Quantity Price  
1           1  Product A      250      20  
2           2  Product B      175      15  
3           3  Product C      300      18  
>  
>  
>
```

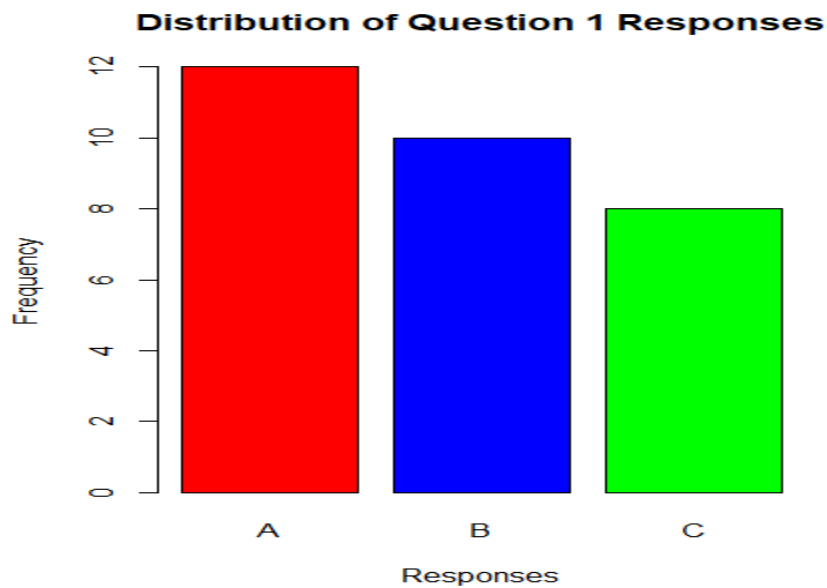
Set 19: Survey Responses Analysis

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

Program:

```
# Survey Responses  
responses <- c("A","B","C")  
counts <- c(12,10,8)  
  
barplot(counts,  
  names.arg = responses,  
  col = c("red","blue","green"),  
  main = "Distribution of Question 1 Responses",  
  xlab = "Responses",  
  ylab = "Frequency")
```

Output:

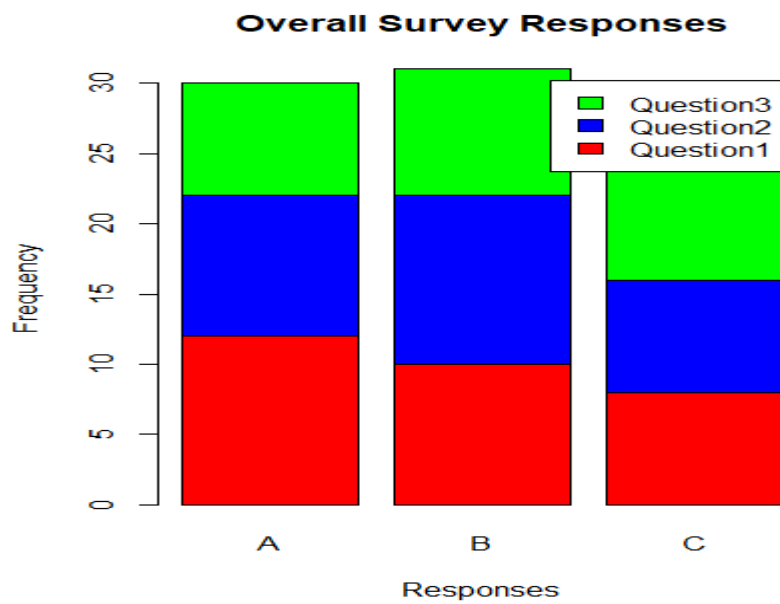


2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

Program:

```
survey <- matrix(c(
  12,10,8,
  10,12,8,
  8,9,13
), nrow=3, byrow=TRUE)
colnames(survey) <- c("A","B","C")
rownames(survey) <- c("Question1","Question2","Question3")
barplot(survey,
  col=c("red","blue","green"),
  main="Overall Survey Responses",
  xlab="Responses",
  ylab="Frequency",
  legend.text=rownames(survey))
```

Output:



3. Build a table to show the survey response data for each survey. Label the table elements.

Program:

```
survey_data <- data.frame(  
  Survey_ID=c(1,2,3),  
  Question1=c("A","B","C"),  
  Question2=c("B","A","A"),  
  Question3=c("C","D","B")  
)
```

survey_data

Output:

```
> source("~/active-rstudio-document")  
> survey_data  
  Survey_ID Question1 Question2 Question3  
1         1         A         B         C  
2         2         B         A         D  
3         3         C         A         B  
> |
```

Set 20: Stock Analysis

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

Program:

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
```

```
stockA <- c(100, 105, 110)
```

```
stockB <- c(150, 152, 148)
```

```
stockC <- c(120, 118, 122)
```

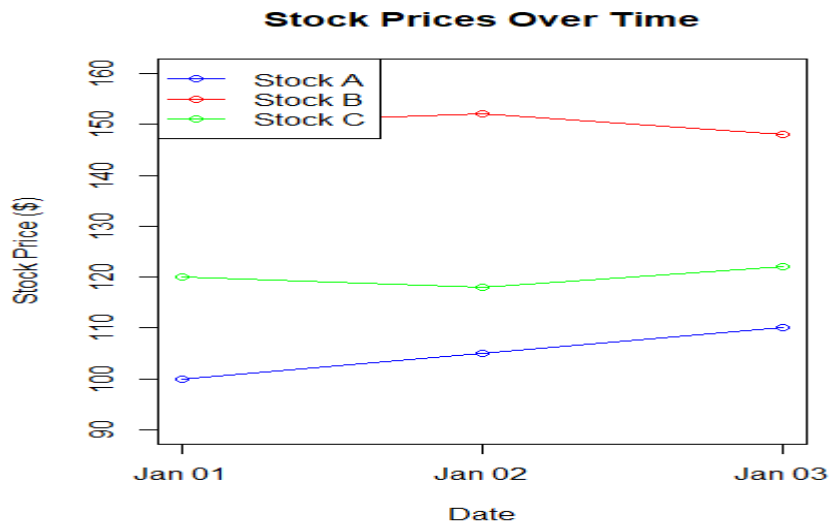
```
plot(date, stockA,  
      type="o",  
      col="blue",  
      ylim=c(90, 160),  
      xlab="Date",  
      ylab="Stock Price ($)",  
      main="Stock Prices Over Time")
```

```
lines(date, stockB,  
      type="o",  
      col="red")
```

```
lines(date, stockC,  
      type="o",  
      col="green")
```

```
legend("topleft",  
      legend=c("Stock A", "Stock B", "Stock C"),  
      col=c("blue", "red", "green"),  
      lty=1,  
      pch=1)
```

Output:

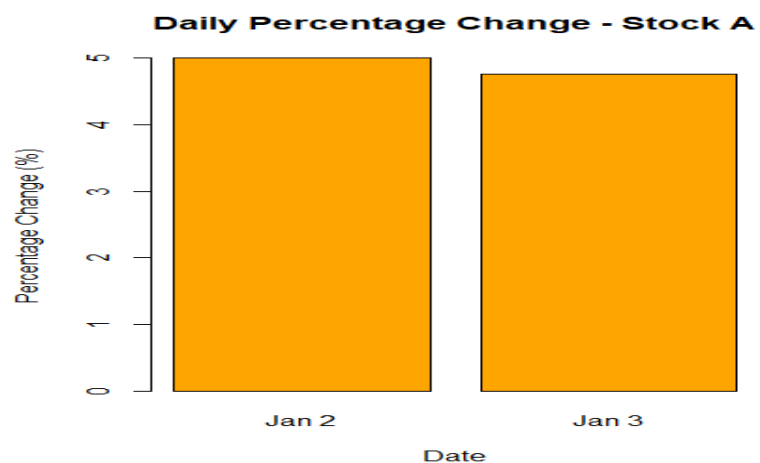


2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

Program:

```
stockA <- c(100,105,110)
percent_change <- diff(stockA)/stockA[-length(stockA)]*100
barplot(percent_change,
        names.arg=c("Jan 2", "Jan 3"),
        col="orange",
        main="Daily Percentage Change - Stock A",
        xlab="Date",
        ylab="Percentage Change (%)")
```

Output:



3. Build a table to display the stock price data for each company over the given period. Label the table elements.

Program:

```
stock_data <- data.frame(  
  Date=c("2023-01-01","2023-01-02","2023-01-03"),  
  Stock_A=c(100,105,110),  
  Stock_B=c(150,152,148),  
  Stock_C=c(120,118,122)  
)  
stock_data
```

Output:

```
< source( "~/ .active-rstudio-document" )  
> source("~/ .active-rstudio-document")  
> stock_data  
      Date Stock_A Stock_B Stock_C  
1 2023-01-01     100     150     120  
2 2023-01-02     105     152     118  
3 2023-01-03     110     148     122  
> |
```